IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 35

BIODIVERSITY- 1

Learning Space

BIODIVERSITY

INTRODUCTION

- Biodiversity is the variety of life on Earth.
- Biodiversity can be studied at three levels -
  - ✓ **Species** - wild plant and animals, domestic animals and crops.
  - ✓ **Ecosystems** – Natural landscapes and Cultural landscapes.
  - ✓ **Genetic Diversity** – within each species of wild/domestic animals and plants.
- So, Biodiversity is the measure of variation at the species, genetic and ecosystem level.
- Biodiversity is the result of 3.5 billion years of evolution.
- Biodiversity is not distributed evenly on Earth, and is richest in the tropics. The tropical forest ecosystems cover less than 10 percent of earth's surface, and contain about 90 percent of the world's species.

Learning Space

BIODIVERSITY

MEGADIVERSITY

- India is one of the 17 mega diverse countries.
- Many of the 17 mega diverse countries are located in, or partially in the tropics and subtropics.
- Mega diversity means exhibiting great diversity. The main criteria for mega diverse countries is endemism, a mega diverse country must have at least 5,000 species of endemic plants.

THE MEGA DIVERSITY COUNTRIES		
Australia	India	Papua New Guinea
Brazil	Indonesia	Peru
China	Madagascar	Philippines
Colombia	Malaysia	South Africa
Ecuador	Mexico	United States
Democratic Republic of the Congo		Venezuela

ECOSYSTEM SERVICES

- These are the benefits people obtain from ecosystems. These are the different goods and services people get freely from the nature.
- There are four categories of Ecosystem Services. They are –
  - ✓ Supporting services
  - ✓ Provisioning services
  - ✓ Regulating services
  - ✓ Cultural services

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ECOSYSTEM SERVICES

1 SUPPORTING SERVICES

- These include services such as
  - ✓ nutrient recycling
  - ✓ primary production
  - ✓ soil formation
  - ✓ pollination etc.
- These services make it possible for the ecosystems to provide other services. So these form the basis for the other services.

ECOSYSTEM SERVICES

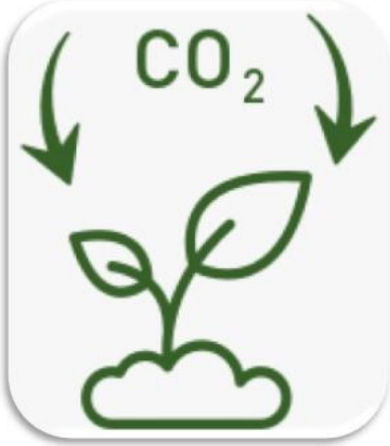
2 PROVISIONING SERVICES

- These services are
  - ✓ **Food** - Food crops, foods such as honey and fruits obtained from the forests.
  - ✓ **Raw materials** - Including timber, skins, fire wood, bio-mass, fodder etc
  - ✓ **Genetic Resources** - Genetic resources are beneficial genes present in the organisms which give the organism beneficial characteristics which are valuable to mankind.
    - ❖ Ex: Food crops which are pest and drought resistant, cattle which give good quality milk etc.
  - ✓ **Fresh Water and Air**
  - ✓ **Medicinal Resources** - Medicinal plants and plat extracts.

ECOSYSTEM SERVICES

3 REGULATING SERVICES

- Carbon Sequestration and climate regulation.
- Waste decomposition
- Purification of water and air.
- Pest control



ECOSYSTEM SERVICES

4 CULTURAL SERVICES

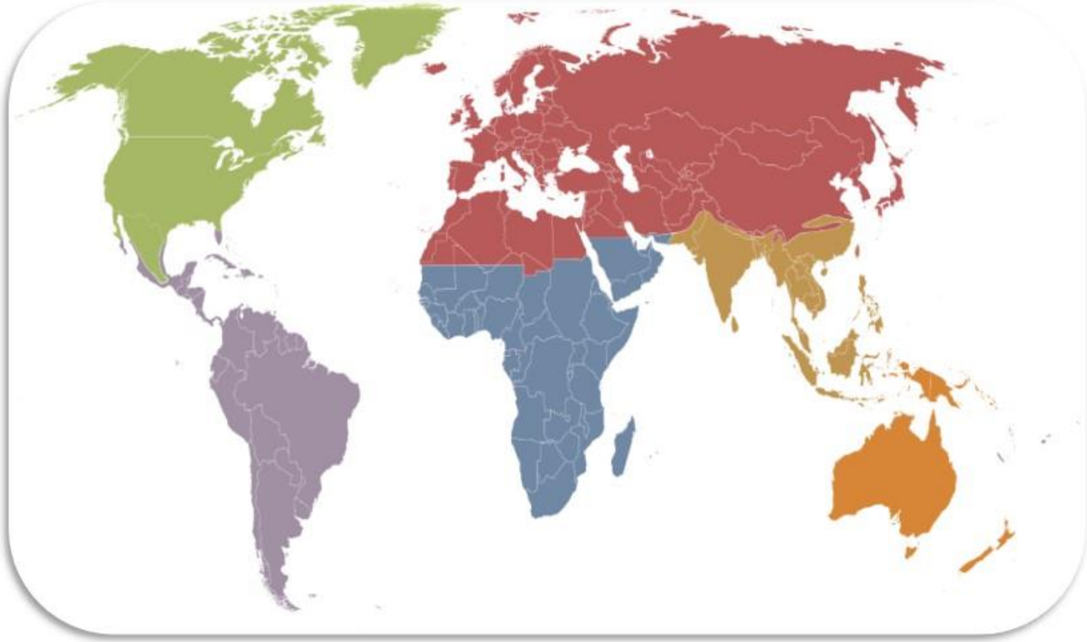
- Use of nature in books, film, painting, folklore, national symbols, advertising etc.
- Spiritual use, including use of nature for religious purposes.
- Recreational use, including eco-tourism, outdoor sports etc.
- Science and education, including use of nature for research, scientific discoveries, learning etc.
- Therapeutic uses, including Eco-therapy, animal-assisted therapy, etc.

KEYSTONE SPECIES

- Any species whose addition or removal from the ecosystem leads to major changes in the populations of several other species.
  - ✓ Honey Bee
- All final consumers are also keystone species.

BIOGEOGRAPHIC REALMS OR ECOZONES

- **Nearctic**
  - ✓ Includes North America
- **Palearctic**
  - ✓ Includes Eurasia and North Africa
- **Afrotropic**
  - ✓ Includes Sub-Saharan Africa
- **Indomalaya**
  - ✓ Includes South Asia and Southeast Asia
- **Australasia**
  - ✓ Includes Australia, New Guinea and neighboring islands.
- **Neotropic**
  - ✓ Includes South America and the Caribbean
- **Oceania**
  - ✓ Includes Polynesia, Fiji and Micronesia.
- **Antarctic**
  - ✓ Includes Antarctica.



BIODIVERSITY HOTSPOT

- The concept of biodiversity hotspots was proposed by Norman Myers in 1988.
- A biodiversity hotspot is a region with significant levels of biodiversity that is threatened with destruction.
- To qualify as a biodiversity hotspot a region must meet two criteria:
  - ✓ It must contain at least 1,500 species of plants as endemics, and
  - ✓ It has to have lost at least 70% of its primary vegetation.
- Around the world, 36 areas qualify under this definition. These sites contain nearly 60% of the world's plant, bird, mammal, reptile, and amphibian species.
- Over 50 percent of the world's plant species and 42 percent of all terrestrial vertebrate species are endemic to the 36 biodiversity hot spots.

NORMAN MYERS

BIODIVERSITY HOTSPOT

- India contains 4 biodiversity hotspots:
  - ✓ The Himalayas
  - ✓ The Western Ghats
  - ✓ The Indo-Burma region and
  - ✓ The Sundalands.
- These hotspots have numerous endemic species.

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